Electrical Engineering Case Study
- Resistor Values

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ENGR-AD 1000, Spring 2023 (Assignment 1)

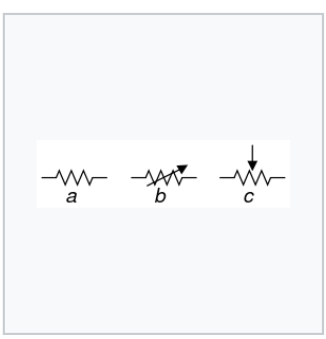
Step 1: Problem Identification and Statement

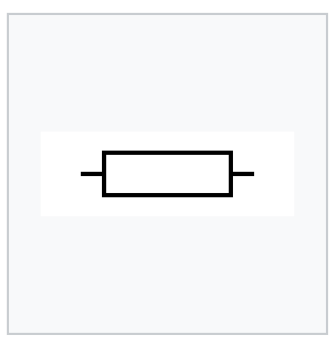
This assignment aims to develop a program that takes input from the user about the colours of the four bands of a resistor and displays its resistance value in kilo-ohms. This is achieved by presenting the user a menu, namely: (1) calculating the resistance value in response to the colours the user inputted, (2) a 'Help' option to present the user with how the program is used, or (3) exiting the program.

Step 2: Gathering Information and Input/Output diagram

- *Gathering of information*

Considering an electrical circuit, a resistor is a circuit element that reduces current flow, adjusts signal levels, divides voltages, and many other functions. The electrical function of a resistor is determined by its resistance, which refers to the amount the resistor impedes an electric current. According to Ohm's Law, the circuit will have a lower current flow if it has a resistor with high resistance. Each resistor has a specific resistance value between its ends, resistance values are measured in ohms (Ω) or kilo-ohms (kΩ). One kilo-ohm refers to 1000 ohms.
Two typical schematic diagram symbols of resistors are as follows:





Resistors have four bands each displaying a color. The first band represents the first significant figure of the resistance value, and the second band represents the second significant figure of the resistance value. The third band represents the power-of-ten multiplier or in other words, the number of zeros. For example, orange represents $\times 10^3$ (a number times 1000). The fourth band

represents the tolerance accuracy or acceptable error. The final resistance value is shown as: 'the resistance value ± tolerance'.

The table below shows the meanings of the colors of the bands.

Table of color bands and corresponding values

Color	1st band	2nd band	3rd band (multiplier)	4th band (tolerance)
Black	***0***	***0***	$\times 10^{0}$	
Brown	***1***	***1***	$\times 10^{1}$	
Red	***2***	***2***	$\times 10^{2}$	
Orange	***3***	***3***	$\times 10^{3}$	
Yellow	***4***	***4***	$\times 10^{4}$	
Green	***5***	***5***	$\times 10^{5}$	
Blue	***6***	***6***	$\times 10^{6}$	
Violet	***7***	***7***	$\times 10^{7}$	
Gray	***8***	***8***	$\times 10^{8}$	
White	***9***	***9***	$\times 10^{9}$	
Gold			$\times 10^{-1}$	***±5%***
Silver			$\times 10^{-2}$	***±10%***

Taking an example.

1. The first band is green, which indicates the first significant figure is 5.
2. The second band is black, which indicates the second significant figure is 0.
3. The third band is orange, which indicates the power-of-ten multiplier to the value is $\times 10^3$. Accompanied by the first two bands the resistance value is $50\times 10^3\Omega$
4. The fourth band is silver, which indicates the tolerance is ±10%. In this case, 10% of $50\times 10^3\Omega$ is $0.1\times 50\times 10^3\Omega$, which is equivalent to 5000Ω.

Overall, it indicates the resistor has a resistance value of $50\times 10^3 \pm 0.1\times 50\times 10^3\Omega$. This can also be written as 50±5kΩ.

With the aid of the table, when the user enters the color of the bands showing on the resistor, the program would be able to display the corresponding resistance value. In this program, the resistance value will be displayed in kilo-ohms, and band 4 is assumed to be only Gold or Silver.

- *Input/Output Description*

Input:

The inputs are the selection of the number in the menu displayed on the screen

1. Input '1' if the user wants to calculate the resistance value,
   - Further input the number for the corresponding color the user sees on the bands. (1) color of band 1, (2) color of band 2, (3) color of band 3, (4) color of band 4. Any input other than the numbers the program assigns to the colors of the bands will be seen as an error.
2. Input '2' for help, when the user wants to see the information about the program and about how to use the program.
3. Input '3' to exit, which terminates the program.
4. Any input other than 1 2 or 3 will be seen as an error.

Outputs:

1. A menu is displayed as output with three options
2. Corresponding to input 1

Display the resistance value of the resistor including the tolerance accuracy. Then display the menu again.

Display an error message if the user did not select the number the program assigned to the colors of the bands. Prompt the user for another value.

3. Corresponding to input 2

Display the information about the program and about how to use the program, a guide for the program. Then display the menu again.

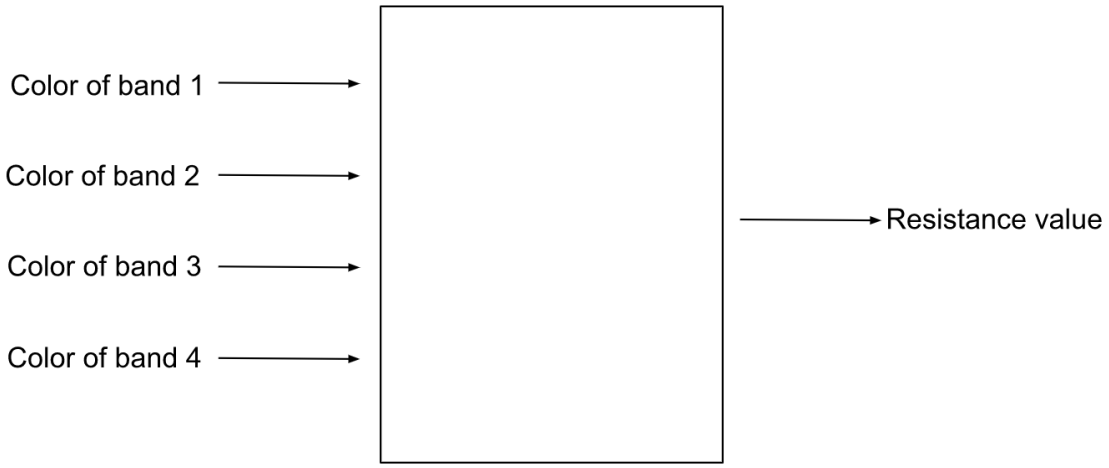
4. Corresponding to input 3

No output, the program is terminated.

5. Corresponding to any other input

An error message will be displayed. Prompt the user for another value.

Inputs and outputs are also shown in the diagram below



Step 3: Test Cases and Algorithm Design

- *Test Cases*

Sample of the main menu

Please choose one of the following: 1) Calculate resistance value 2) Help 3) Exit

Test Case 1: Invalid input in menu selection

In the menu, users select the function they wish the program to. If the user enters an invalid input (numbers other than 1, 2, or 3), the program must first display an error message and then prompt the user for another valid value.

Test Case 2: Selecting 1) Calculate resistance value

If the user selected 1) Calculate resistance value, the program must display each band in the order from 1 to 4. For each band, all colors of the band are assigned by the program a unique number for the user to select according to the table below.
For bands 1 and 2, 1) to 10) will be displayed.
For band 3, all twelve options will be displayed.
For band 4, only 11) Gold and 12) Silver will be displayed.

Color	Black	Brown	Red	Orange	Yellow	Green	Blue	Violet	Grey	White	Gold	Silver
Number	1	2	3	4	5	6	7	8	9	10	11	12

After the user has chosen colors for all the bands, the program must display the corresponding resistance value of the resistor in kilo-ohms. The resistance value is calculated according to the *Table of color bands and corresponding values* mentioned in "Gathering information". The user will then be automatically directed back to the menu for re-selection.
The following test cases will be performed in "Test and Verification"
Combination 1) White, grey, brown, gold. Should return 0.98±0.049kΩ
2) Blue, orange, green, gold. Should return 6300±315kΩ
The user will then be automatically directed back to the menu for re-selection.

Test Case 3: Invalid input for band 1

If the user's input for band 1 is not one of the numbers assigned by the program to the colors of band 1, i.e., smaller than 1 or larger than 10, the program must notify the user it is invalid and prompt for another value.
The following test cases will be performed in "Test and Verification"

1) Input invalid value -1

2) Input invalid value 100

Note: the user is assumed to input only numerical integer values.

Test Case 4: Invalid input for band 2

If the user's input for band 2 is not one of the numbers assigned by the program to the colors of band 1, i.e., smaller than 1 or larger than 10, the program must notify the user it is invalid and prompt for another value.

The following test cases will be performed in "Test and Verification"

1) Input invalid value -2
2) Input invalid value 200

Note: the user is assumed to input only numerical integer values.

Test Case 5: Invalid input for band 3

If the user's input for band 3 is not one of the numbers assigned by the program to the colors of band 1, i.e., smaller than 1 or larger than 12, the program must notify the user it is invalid and prompt for another value.

The following test cases will be performed in "Test and Verification"

1) Input invalid value -3
2) Input invalid value 300

Note: the user is assumed to input only numerical integer values.

Test Case 6: Invalid input for band 4

If the user's input for band 4 is not one of the numbers assigned by the program to the colors of band 1, i.e., smaller than 11 or larger than 12, the program must notify the user it is invalid and prompt for another value.

The following test cases will be performed in "Test and Verification"

1) Input invalid value -4
2) Input invalid value 400

Note: the user is assumed to input only numerical integer values.

Test Case 7: Selecting 2) Help

If the user selected 2) Help, the program must display the information about this program and instructions on how this program is used. After the user has acknowledged it, the user will be automatically directed back to the menu and re-select the function the user wants to perform.

Test Case 8: Selecting 3) Exit

If the user selected 3) Exit, the program must display a message, such as "Thank you for using the program, goodbye!". And then quit the program.

- *Algorithm Design*

Declare choice, band1 (initialized to 0), band2 (initialized to 0), band3 (initialized to 0), band4(initialized to 0)
Declare resistance, resistancek, tolerance, num1 (initialized to -1), num2 (initialized to -1), num3 (initialized to -1), num4 (initialized to -1)

Repeat the following while choice≠3
    Print newline
    Print "Please enter the corresponding number to the function you wish to perform", newline, "1) Calculate resistance value", newline, "2) Help", newline, "3) Exit", newline
    Print newline
    Read value into choice

    Repeat while choice<1 or choice>3
        Print "Invalid input, please select a number between 1, 2, and 3", newline
        Read value into choice

    If choice=1
        Print newline
        Print "Please enter the corresponding number to the color you see for band 1 on the resistor", newline, "1) Black", newline, "2) Brown", newline, "3) Red", newline, "4) Orange", newline, "5) Yellow", newline, "6) Green", newline, "7) Blue", newline, "8) Violet", newline, "9) Grey", newline, "10) White", newline
        Read value into band1
        Repeat while band1<1 or band1>10
            Print "Invalid input, please enter the corresponding number to the color you see for band 1 on the resistor", newline
            Read value into band1
        Assign band1-1 to num1

        Print newline
        Print "Please enter the corresponding number to the color you see for band 2 on the resistor", newline, "1) Black", newline, "2) Brown", newline, "3) Red", newline, "4) Orange", newline, "5) Yellow", newline, "6) Green", newline, "7) Blue", newline, "8) Violet", newline, "9) Grey", newline, "10) White", newline
        Read value into band2
        Repeat while band2<1 or band2>10
            Print "Invalid input, please enter the corresponding number to the color you see for band 2 on the resistor", newline

```
        Read value into band2
Assign band2-1 to num2

Print newline
Print "Please enter the corresponding number to the color you see for band 3 on the
resistor", newline, "1) Black", newline, "2) Brown", newline, "3) Red", newline, "4)
Orange", newline, "5) Yellow", newline, "6) Green", newline, "7) Blue", newline, "8)
Violet", newline, "9) Grey", newline, "10) White", newline, "11) Gold", newline, "12)
Silver", newline
Read value into band3
Repeat while band3<1 or band3>12
        Print "Invalid input, please enter the corresponding number to the color
        you see for band 3 on the resistor", newline
        Read value into band3
If Band3≤10
   Assign 10 to the power of band3-1 to num3
Otherwise if Band3=11
        Assign 10 to the power of -1 to num3
Otherwise if Band3=12
        Assign 10 to the power of -2 to num3

Print newline
Print "Please enter the corresponding number to the color you see for band 4 on the
resistor", newline, "11) Gold", newline, "12) Silver", newline
Read value into band4
Repeat while band4<11 or band4>12
        Print "Invalid input, please enter the corresponding number to the color
        you see for band 4 on the resistor", newline
        Read value into band4
If Band4=11
   Assign 0.05 to num4
Otherwise if Band4=12
        Assign 0.1 to num4

Assign num1*(num3*10)+num2*num3 to resistance
Assign resistance/1000 to resistancek
Assign resistancek*num4 to tolerance

Print newline
Print "The resistance value is:", resistancek, "±", tolerance, "kΩ", newline
```

Print newline
Continue

If choice=2

Print newline
Print “This program is to help you obtain the resistor's resistance value. To use it, the colors on the four bands shown on the resistor are associated with a number which will be displayed to you, simply select the number of the color you see for each band as instructed by the program step by step.”, newline
Print newline
Continue

If choice=3

Print newline
Print “Thank you for using the program, goodbye!”
Exit

Step 4: Code of implementation

```
/*--------------------------------*/
/*Name: Zichen Jing (Nancy), NetID: zj2256 */
/*        Date: March 5, 2023        */
/*         Assignment 1.cpp         */
/*--------------------------------*/

/*----------------------------------------START-----------------------------------
---------*/

#include <iostream>
#include <iomanip>
#include <cmath>

using namespace std;

int main() {
    //declare variables, initialized to values that will never naturally appear in
    program.
    int choice, band1(0), band2(0), band3(0), band4(0); //choice corresponds to the
        choice of menu, band corresponds to the number the user chose for each band
    double resistance, resistancek, tolerance, num1(-1), num2(-1), num3(-1), num4(-1);
        //resistancek corresponds to the resistance value in kilo-ohms, num
        corresponds to the value represented by the user's selection for each band

    do{
        //display menu to the user
        cout<<"\n"<<endl; //empty line to make the statements displayed less
                                crowded
        cout<<"Please enter the corresponding number to the function you wish to
            perform"<<endl;
        cout<<"1) Calculate resistance value"<<endl;
        cout<<"2) Help"<<endl;
```

```
cout<<"3) Exit"<<endl;
cout<<"\n"<<endl; //empty line to make the statements displayed less
                        crowded
cin>>choice;

//check if the user's input is valid, must be one of the numbers provided
while (choice<1||choice>3){
      cout<<"Invalid input, please select a number between 1, 2, and 3"<<endl;
      cin>>choice;}

//three cases created using switch structure depend on the user's choice
switch (choice){
    //when the user chose to let the program calculate the resistance value
    case 1:
        cout<<"\n"<<endl; //empty line to make the statements displayed less
                        crowded
        cout<<"Please enter the corresponding number to the color you see for
            band 1 on the resistor"<<endl;
        //display the numbers the program associated with the colors of band 1
        for the user to choose
        cout<<"1) Black"<<endl;
        cout<<"2) Brown"<<endl;
        cout<<"3) Red"<<endl;
        cout<<"4) Orange"<<endl;
        cout<<"5) Yellow"<<endl;
        cout<<"6) Green"<<endl;
        cout<<"7) Blue"<<endl;
        cout<<"8) Violet"<<endl;
        cout<<"9) Grey"<<endl;
        cout<<"10) White"<<endl;
        cin>>band1; //storing the selection to variable band1

        //check if the user's input is valid, must be one of the numbers
        provided
        while (band1<1||band1>10){
            cout<<"Invalid input, please enter the corresponding number to the
                color you see for band 1 on the resistor"<<endl;
            cin>>band1;}

        num1=band1-1; //according to the diagram given, for band 1, the number
            the program associated with the color starts at 1 whereas the
            significant digit corresponding to the color starts at 0,
            therefore the significant digit(num1) is always one less than the
            number program assigned to the color(band1)

        cout<<"\n"<<endl; //empty line to make the statements displayed less
                        crowded
        cout<<"Please enter the corresponding number to the color you see for
            band 2 on the resistor"<<endl;
        //display the numbers the program associated with the colors of band 2
        for the user to choose
        cout<<"1) Black"<<endl;
        cout<<"2) Brown"<<endl;
        cout<<"3) Red"<<endl;
        cout<<"4) Orange"<<endl;
        cout<<"5) Yellow"<<endl;
        cout<<"6) Green"<<endl;
        cout<<"7) Blue"<<endl;
        cout<<"8) Violet"<<endl;
        cout<<"9) Grey"<<endl;
        cout<<"10) White"<<endl;
```

```
cin>>band2; //storing the selection to variable band2

//check if the user's input is valid, must be one of the numbers
provided
while (band2<1||band2>10){
    cout<<"Invalid input, please enter the corresponding number to the
        color you see for band 2 on the resistor"<<endl;
    cin>>band2;}

num2=band2-1; //According to the diagram given, for band 2, the number
    the program associated with the color starts at 1 whereas the
    significant digit corresponding to the color starts at 0,
    therefore the significant digit(num2) is always one less than the
    number program assigned to the color(band2)

cout<<"\n"<<endl; //empty line to make the statements displayed less
                  crowded
cout<<"Please enter the corresponding number to the color you see for
     band 3 on the resistor"<<endl;
//display the numbers the program associated with the colors of band 3
for the user to choose
cout<<"1) Black"<<endl;
cout<<"2) Brown"<<endl;
cout<<"3) Red"<<endl;
cout<<"4) Orange"<<endl;
cout<<"5) Yellow"<<endl;
cout<<"6) Green"<<endl;
cout<<"7) Blue"<<endl;
cout<<"8) Violet"<<endl;
cout<<"9) Grey"<<endl;
cout<<"10) White"<<endl;
cout<<"11) Gold"<<endl;
cout<<"12) Silver"<<endl;
cin>>band3; //storing the selection to variable band3

//check if the user's input is valid, must be one of the numbers
  provided
while (band3<1||band3>12){
    cout<<"Invalid input, please enter the corresponding number to the
        color you see for band 3 on the resistor"<<endl;
    cin>>band3;}
//according to the diagram given, for band 3, the number program
associated to the color starts at 1 whereas the multiplier
corresponding to the color starts at 10 to the power of 0, therefore
including 10 to the power of 9, below it, the power in the
multiplier(num3) is always one less than the number program assigned
to the color(band3)
if (band3<=10)
    num3=pow(10,(band3-1));
//for 10 to the power of a negative value, listed separately as it
does not follow the pattern stated above
else if (band3==11)
    num3=pow(10,-1);
else if (band3==12)
    num3=pow(10,-2);

cout<<"\n"<<endl; //empty line to make the statements displayed less
                  crowded
cout<<"Please enter the corresponding number to the color you see for
    band 4 on the resistor"<<endl;
//display the numbers the program associated with the colors of band 4
```

```
            for the user to choose, assume band 4 can only be Gold or Silver
            cout<<"11) Gold"<<endl;
            cout<<"12) Silver"<<endl;
            cin>>band4; //storing the selection to variable band4

            //check if the user's input is valid, must be one of the numbers
            provided
            while (band4<11||band4>12){
                cout<<"Invalid input, please enter the corresponding number to the
                    color you see for band 4 on the resistor"<<endl;
                cin>>band4;}
            if (band4==11)
                //5% converted to 0.05 to have easier calculations in the program
                num4=0.05;
            else if (band4==12)
                //10% converted to 0.1 to have easier calculations in the program
                num4=0.1;

            //resistance calculated in ohms
            resistance=num1*(num3*10)+num2*num3;
            //resistance calculated in kilo-ohms
            resistancek=resistance/1000;
            //tolerance calculated in kilo-ohms
            tolerance=resistancek*num4;

            //print both resistance value and tolerance in kilo-ohms,
            including '±' symbol and the unit kilo-ohms kΩ
            cout<<"\n"<<endl; //empty line to make the statements displayed less
                              crowded
            cout<<"The resistance value is: " <<resistancek<<"±"<<tolerance<<"kΩ"
                <<endl;
            cout<<"\n"<<endl;
            //allows looping back to the menu
            continue;

        //when the user chose to see the instruction of the program
        case 2:
            cout<<"\n"<<endl; //empty line to make the statements displayed less
                              crowded
            cout<<"This program is to help you obtain the resistor's resistance
                value. To use it, the colors on the four bands shown on the
                resistor are associated with a number which will be displayed to
                you, simply select the number of the color you see for each band
                as instructed by the program step by step."<<endl;
            cout<<"\n"<<endl;
            //allows looping back to the menu
            continue;

        //when the user chose to exit, a message will be printed, and then exit
        the program
        case 3:
            cout<<"\n"<<endl;//empty line to make the statements displayed less
                             crowded
            cout<<"Thank you for using the program, goodbye!"<<endl;
            exit(1); //exit program will receive a return value of 1
    }

}while (choice!=3); //can only exit the program by choosing 3)Exit from the menu,
    for case 1 and case 2, once finished, the menu will be automatically re-
    displayed
```

```
    return 0;
}

/*-----------------------------------END-----------------------------------------*/
```

Step 5: Test and Verification

Menu display

```
Please enter the corresponding number to the function you wish to perform
1) Calculate resistance value
2) Help
3) Exit
```

Test Case 1: Invalid input in menu selection

```
Please enter the corresponding number to the function you wish to perform
1) Calculate resistance value
2) Help
3) Exit


4
Invalid input, please select a number between 1, 2, and 3
```

(User's input is 4, which is invalid, an error message is displayed, and asked the user for another value.)

Test Case 2: Selecting 1) Calculate resistance value

```
Please enter the corresponding number to the function you wish to perform
1) Calculate resistance value
2) Help
3) Exit


1


Please enter the corresponding number to the color you see for band 1 on the resistor
1) Black
2) Brown
3) Red
4) Orange
5) Yellow
6) Green
7) Blue
8) Violet
9) Grey
10) White
10


Please enter the corresponding number to the color you see for band 2 on the resistor
1) Black
2) Brown
3) Red
4) Orange
5) Yellow
6) Green
7) Blue
8) Violet
9) Grey
10) White
9
```

```
Please enter the corresponding number to the color you see for band 3 on the resistor
1) Black
2) Brown
3) Red
4) Orange
5) Yellow
6) Green
7) Blue
8) Violet
9) Grey
10) White
11) Gold
12) Silver
2


Please enter the corresponding number to the color you see for band 4 on the resistor
11) Gold
12) Silver
11


The resistance value is: 0.98±0.049kΩ




Please enter the corresponding number to the function you wish to perform
1) Calculate resistance value
2) Help
3) Exit
```

(User's input is 1 to calculate resistance value. Colors for four bands are displayed with the correct number assigned in the correct order for the user to choose. The first test case is white, grey, brown, and gold. The correct resistance value of 0.98±0.049kΩ was displayed with unit kilo-ohms. Automatically directed back to the menu after the calculation.)

```
Please enter the corresponding number to the function you wish to perform
1) Calculate resistance value
2) Help
3) Exit


1


Please enter the corresponding number to the color you see for band 1 on the resistor
1) Black
2) Brown
3) Red
4) Orange
5) Yellow
6) Green
7) Blue
8) Violet
9) Grey
10) White
7


Please enter the corresponding number to the color you see for band 2 on the resistor
1) Black
2) Brown
3) Red
4) Orange
5) Yellow
6) Green
7) Blue
8) Violet
9) Grey
10) White
4
```

```
Please enter the corresponding number to the color you see for band 3 on the resistor
1) Black
2) Brown
3) Red
4) Orange
5) Yellow
6) Green
7) Blue
8) Violet
9) Grey
10) White
11) Gold
12) Silver
6


Please enter the corresponding number to the color you see for band 4 on the resistor
11) Gold
12) Silver
11


The resistance value is: 6300±315kΩ




Please enter the corresponding number to the function you wish to perform
1) Calculate resistance value
2) Help
3) Exit
```

(The second test case is blue, orange, green, and gold. The correct resistance value of 6300±315kΩ was displayed with unit kilo-ohms. Automatically directed back to the menu after the calculation.)

Test Case 3: Invalid input for band 1

```
Please enter the corresponding number to the color you see for band 1 on the resistor
1) Black
2) Brown
3) Red
4) Orange
5) Yellow
6) Green
7) Blue
8) Violet
9) Grey
10) White
-1
Invalid input, please enter the corresponding number to the color you see for band 1 on the resistor
100
Invalid input, please enter the corresponding number to the color you see for band 1 on the resistor
```

(User's input is -1 and 100, which are invalid, an error message is displayed and asked the user for another valid input.)

Test Case 4: Invalid input for band 2

```
Please enter the corresponding number to the color you see for band 2 on the resistor
1) Black
2) Brown
3) Red
4) Orange
5) Yellow
6) Green
7) Blue
8) Violet
9) Grey
10) White
-2
Invalid input, please enter the corresponding number to the color you see for band 2 on the resistor
200
Invalid input, please enter the corresponding number to the color you see for band 2 on the resistor
```

(User's input is -2 and 200, which are invalid, an error message is displayed and asked the user for another valid input.)

Test Case 5: Invalid input for band 3

```
Please enter the corresponding number to the color you see for band 3 on the resistor
1) Black
2) Brown
3) Red
4) Orange
5) Yellow
6) Green
7) Blue
8) Violet
9) Grey
10) White
11) Gold
12) Silver
-3
Invalid input, please enter the corresponding number to the color you see for band 3 on the resistor
300
Invalid input, please enter the corresponding number to the color you see for band 3 on the resistor
```

(User's input is -3 and 300, which are invalid, an error message is displayed and asked the user for another valid input.)

Test Case 6: Invalid input for band 4

```
Please enter the corresponding number to the color you see for band 4 on the resistor
11) Gold
12) Silver
-4
Invalid input, please enter the corresponding number to the color you see for band 4 on the resistor
400
Invalid input, please enter the corresponding number to the color you see for band 4 on the resistor
```

(User's input is -4 and 400, which are invalid, an error message is displayed and asked the user for another valid input.)

Test Case 7: Selecting 2) Help

```
Please enter the corresponding number to the function you wish to perform
1) Calculate resistance value
2) Help
3) Exit


2


This program is to help you obtain the resistor's resistance value. To use it, the colors on the four bands shown on the
    resistor are associated with a number which will be displayed to you, simply select the number of the color you see
    for each band as instructed by the program step by step.




Please enter the corresponding number to the function you wish to perform
1) Calculate resistance value
2) Help
3) Exit
```

(User's input is 2 for help. The information about the program and instructions to use the program is correctly displayed. Automatically directed back to the menu after the calculation.)

Test Case 8: Selecting 3) Exit

```
Please enter the corresponding number to the function you wish to perform
1) Calculate resistance value
2) Help
3) Exit


3


Thank you for using the program, goodbye!
Program ended with exit code: 1
```

(User's input is 3 for exit. Message correctly displayed and the program terminates.)

The User's Guide

In the beginning, a menu will be displayed to you. You can make one of three choices depending on what you would like the program to do. Please enter your selection as number 1, 2, or 3 and press Enter.

If you choose 1, you will be then asked to enter the number assigned by the program to the colors you see on each band on the resistor. Any invalid input will result in the program displaying an error message and asking you for another valid input. Once all fields are correctly entered, the program will then calculate the resistance value and tolerance of the resistor based on your inputs of the colors the resistor displayed. The program will then automatically direct you back to the menu for another calculation, instructions about the program, or exit.

If you choose 2, information about this program and how to use the program will be displayed as instructions for you. The program will then automatically direct you back to the menu to re-select the function you want the program to perform.

If you choose 3, the screen will display a message and then the program will end.